THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM THREE
033 BIOLOGY

Time: 3 Hours November 2024

Instructions
1. This paper consists of sections A, B and C with a total of fifteen (11) questions.
2. Answer all questions in sections A, B, and two (2) questions from section C.
4. All writing should be in blue ink, except for diagrams, which must be drawn in pencil.
5. Cellular phones and any unauthorized materials are not allowed in the examination room.
7. Write your Examination Number on every page of your answer sheets.

SECTION A (16 Marks)
Answer all questions in this section

1. Choose the most correct answer among the given alternatives and write down the answer in the space given.

(i) What type of joint is formed by the atlas and the axis?.
A. Hinge joint B. pivot joint C. Glide joint
D. Ball and socket joint E. Saddle joint

(ii) Which of the following is set of raw materials and conditions required for photosynthesis to take place?
A. Chlorophyll, sunlight, oxygen and water
B. Chlorophyll, sunlight, carbon dioxide and water
C. Chlorophyll, sunlight, nitrogen and water
D. Chlorophyll, sunlight, carbon dioxide and oxygen
E. Sunlight, carbon dioxide, water and oxygen

(iii) What is the functional role a species plays in an ecological system with respect to the species habitat requirement called?
A. Ecological niche
B. Trophic level
C. Interspecific competition
D. Intraspecific competition
E. Perpetual ecosystem

(iv) Which of the following is true about malaria?
A. I t is caused by bacteria
B. It is caused by virus

C. It is caused by a protozoa
D. It is caused by fungus
E. It is caused by mosquitoes

(v) Consider a fruit such as the orange, what part of its flower develops into its seeds?
A. Ovary
B. Stigma
C. Ovule
D. Style
E. Anther

(vi) An experiment lacking the tested factor only is called
A. Independent experiment
B. Dependent experiment
C. Hypothesis
D. Conclusion
E. Control experiment

(vii) Which of the following are correct about the likely cause of diabetes mellitus?
A. Cells inability to utilize insulin or too much production of insulin in the body
B. Cells inability to utilize insulin or too little production of insulin in the body
C. Cells inability to utilize glucagon or too little production of glucagon in the body
D. Too much insulin in the blood with too little blood sugar
E. Too little blood sugar with too much glucose

(viii) Which of the following is present in animal cell but absent in the higher plants cell?
A. Nucleus
B. Ribosome
C. Centriole
D. Mitochondria
E. Chlorophyll

(ix) Which of the following has the function of transmitting sound vibrations from the eardrum to the oval window in human ear?
A. Hammer, anvil and ampulla
B. Semicircular canals, tympanum and stirrups
C. Stirrups, malleus and Eustachian tube
D. Malleus, incus and stapes
E. Malleus, sacculus and utriculus

(x) Seypalm drinks too much alcohol, which one of his internal organ is overworked and could be damaged?
A. Liver
B. Pancreases
C. Heart

D. Lungs
E. Eye

2. Match items in list A with the correct phrase in list B

LIST A		LIST B
(i)	They show alternation of generations and the gametophyte generation is dominant	A. Filicinophyta B. Coniferophyta C. Bryophytes D. Zygomycota E. Basidiomycota F. Rhizoids G. Rhizome H. Monocotyledonous I. Dicotyledonous J. Angiospermophyta
(ii)	They show alternation of generations and the gametophyte is a free-living organism.	
(iii)	They grow on rotten materials such as bread, and jam, where they appear as mass of fine threads and small black dots	
(iv)	The reproductive part (fruiting body) is found above the ground and the hyphae or mycelium is located under the ground	
(v)	Floral parts occur in fours and fives and multiple of four or five	
(vi)	Hair like structures that absorb nutrients for the moss plant.	

SECTION B (54 Marks)

Answer all questions in this section.

3. (a) Briefly explain how burns and scalds are different.
(b) Two students were involves in accidents in a Biology laboratory during an experiment. One of them suffered burns on his hand from the flame of the Bunsen burner, while the other one, had concentrated sulphuric acid spilled on her hand. Outline the first aid procedures you would follow when helping each of them.

4. (a) Differentiate between nutrition and nutrients.
(b). Heterotrophic nutrition can be divided into three major categories. Briefly explain what it is and give an account of each of its categories.

5. Rate of breathing of a normal adult human at rest is 12 to 18 breaths per minute. Explain six factors that can alter this rate.

6. (a) Briefly explain three differences between the skeletal and cardiac muscles.
(b) Giving at least three points, briefly explain the functions of the human skeleton.

7. **(**a) Panting in dogs is likened to sweating in humans. Compare and contrast the two processes with respect to adjusting over-heating in both dogs and humans.
(b)Outline the mechanism of osmoregulation and blood sugar regulation in humans.

8. (a) Outline the five components of nervous coordination.
(b) Draw a diagram of a typical neurone

SECTION C (30 Marks)

Answer two questions from this section

9. Drug Abuse and Addiction and addiction is globally a serious problem. Write an article to educate the public about the problem with focus on the types of drug addiction, effects of drug abuse and how it can be controlled.

10. You're invited by the district councilor from one district from a certain region as a Biologist with knowledge about Tuberculosis. Explain how you will educate the dwellers of the district about tuberculosis on the cause, sign/ symptoms, control and treatments.

11. Write an essay on blood transfusion using the following guidelines:
    (i) Meaning of blood transfusion
    (ii) Types of blood groups
    (iv) Importance of blood transfusion
    (iii) Precautions taken to ensure safety of recipient